

Capstone Project: Azure DevOps CI Pipeline for a Banking Loan Eligibility App

Business Scenario

A bank wants a simple Python-based loan eligibility application. The application should check whether a customer is eligible for a loan based on salary, credit score, and employment status.

The engineering team must plan the work in Azure DevOps, store the Python code in Azure Repos, and create a CI pipeline to run the application automatically.

Part 1 – Agile Planning

Epic

Develop a Loan Eligibility System for a Banking Application

The system should evaluate customer loan eligibility using simple business rules.

Your Task

Break this Epic into:

```
Features
User Stories
Tasks
```

Create the hierarchy in Azure DevOps Boards.

Do not create only one flat list. Use the structure:

```
Epic
↓
```

Feature



User Story



Task

Part 2 – Python Application

Create an Azure DevOps repository named:

```
loan-eligibility-system
```

Create a Python file:

```
loan_app.py
```

Sample code:

```
class LoanEligibility:

    def __init__(self, customer_name, salary, credit_score, employed):
        self.customer_name = customer_name
        self.salary = salary
        self.credit_score = credit_score
        self.employed = employed

    def check_eligibility(self):
        if self.salary >= 50000 and self.credit_score >= 700 and self.employed:
            return "Eligible for Loan"
        else:
            return "Not Eligible for Loan"

customer = LoanEligibility(
    "Priya Reddy",
    65000,
    750,
    True
)
```

```
print("Customer Name:", customer.customer_name)
print("Loan Status:", customer.check_eligibility())
```

Expected output:

```
Customer Name: Priya Reddy
Loan Status: Eligible for Loan
```

Part 3 – CI Pipeline

Create:

```
azure-pipelines.yml

trigger:
- main

pool:
  vmImage: ubuntu-latest

steps:
- task: UsePythonVersion@0
  inputs:
    versionSpec: '3.x'

- script: |
    python loan_app.py
  displayName: 'Run Loan Eligibility App'
```

Part 4 – Validation

Participants must verify:

1. Epic is created.
2. Features are created under Epic.

3. User Stories are created under Features.
 4. Tasks are created under User Stories.
 5. Code is pushed to Azure Repos.
 6. Pipeline is created.
 7. Pipeline runs successfully.
 8. Output is visible in logs.
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Deliverables

Document with Screenshots of the below.

Azure DevOps Board
Azure Repo
loan_app.py
azure-pipelines.yml
Successful pipeline run screenshot